Graphon Mean Field Games with Control Constraints Dependent upon Network Control Mean Fields

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***Abstract*— In many real-world game-theoretic scenarios, the resources available to individual agents are subject to strict limitations. This work investigates such dynamics within the framework of Extended Graphon Mean Field Games (EGMFG), which is governed by a coupled system of Hamilton-Jacobi-Bellman (HJB) and Fokker-Planck-Kolmogorov (FPK) equations. We model these resource limitations as hard constraints on each agent's control variable, continuously depending on the local network control mean field, which serves as a macroscopic proxy for the local resource level. By establishing the Lipschitz and Hölder continuity of the minimized Hamiltonian under Hausdorff-continuous moving constraints, we guarantee the existence of classical solutions to the associated HJB equation. Subsequently, the existence of a globally consistent joint probability measure flow is obtained via a contraction mapping approach. Finally, numerical simulations incorporating both time-varying and universal constraints are presented, illustrating the saturated behavior of the optimal control policies.**

I. INTRODUCTION

Mean Field Games (MFGs) provide a powerful framework for analyzing differential games involving an infinite population of interacting agents. In realistic settings, agents frequently operate under strict constraints on their states or controls. However, incorporating these constraints introduces significant mathematical challenges. For instance, preference controls induced by cost functions may drive states toward the boundary of feasible regions, causing the probability measure of the state or control to lose absolute continuity. The resulting emergence of singular measures on boundaries often renders standard partial differential equation (PDE) approaches intractable. Consequently, analyzing the existence of a Lagrangian equilibrium—represented directly by a probability measure flow over state and control trajectories—has become a prevalent methodology in constrained MFG problems, as seen in [1], [2], [3], [4].

Existing literature has explored MFGs with constraints, particularly in deterministic settings. For example, crowd motion models restrict movement within specific spatial domains, utilizing accelerations as controls and incorporating congestion terms into the cost function [3]. To ensure well-posedness, regularity conditions on the initial state distributions and cost functions are typically proposed to guarantee that optimal trajectories do not encounter infeasible limiting directions. In such formulations, topological fixed-point theorems, such as Kakutani's theorem, are employed to prove the existence of a Lagrangian equilibrium [4].

Despite the success of direct Lagrangian formulations, the most straightforward approach remains solving the coupled Hamilton-Jacobi-Bellman (HJB) and Fokker-Planck-Kolmogorov (FPK) system to construct a contraction mapping. This methodology has been successfully extended to Graphon Mean Field Games (GMFGs) [5], [6], [7], which model heterogeneous, dense interactions over large-scale networks. In this work, we advance this paradigm by studying an Extended Graphon Mean Field Game (EGMFG) where the constraints on an agent's control depend on the local network control mean field. This dynamic, distribution-dependent constraint fundamentally couples the feasible action space of an individual agent to the collective control behavior of its network neighbors, a feature that necessitates analyzing the joint probability measure of states and controls.

To tackle this coupled system, we assume the constrained sets for the controls are sufficiently regular to maintain Lipschitz and Hölder continuity. These are reasonable assumptions that ensure the existence of classical solutions to the associated HJB equations. To establish the existence and uniqueness of the Nash equilibrium, we construct a fixed-point mapping on the complete space of path measure ensembles. By evaluating the sensitivity of the generalized McKean-Vlasov dynamics to the induced optimal policies, we prove that this mapping is a contraction under a weak coupling condition, thereby uniquely determining the consistent joint measure flow. Finally, we provide a numerical scheme and MATLAB simulations that illustrate the theoretical findings, highlighting the distinct macroscopic behaviors induced by mean-field-dependent control constraints.

II. PRELIMINARIES

A. Notations

This paper aims at solving the following HJB-FPK system of a GMFG with control constraints depend on the network control mean fields.

$$
\begin{aligned}
&[HJB](\alpha) \quad -\frac{\partial V^\alpha(t,x)}{\partial t} = \inf_{+u\in U_\alpha(t,\bar{\lambda}_\alpha)} \Big\{ \tilde{l}(x,u,\boldsymbol{\nu}_G; g_\alpha) \\
&\quad \tilde{f}(x,u,\boldsymbol{\nu}_G; g_\alpha)\frac{\partial V^\alpha(t,x)}{\partial x} \Big\} + \frac{\sigma^2}{2}\frac{\partial^2 V^\alpha(t,x)}{\partial x^2}, \\
&[FPK](\alpha) \quad \frac{\partial p_\alpha(t,x)}{\partial t} = \frac{-\partial\big\{\tilde{f}(x,u^o,\boldsymbol{\nu}_G; g_\alpha) p_\alpha(t,x)\big\}}{\partial x} \\
&\quad + \frac{\sigma^2}{2}\frac{\partial^2 p_\alpha(t,x)}{\partial x^2}, \\
&[BR](\alpha) \quad u^0 := \varphi(t,x,\boldsymbol{\nu}_G; g_\alpha),
\end{aligned} \tag{1}
$$

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where $\boldsymbol{\nu}_G \in C([0,1], \mathcal{P}(\mathbb{R}^n \times \mathbb{R}^n))$ is the joint probability measure ensemble of state and control at a specific time, $\overline{\lambda}_\alpha$ is the network state and control mean field that will be defined in (6), $U_\alpha(t, \overline{\lambda}_\alpha)$ is the time wise feasible control set that will be defined in (4), and the functions $\tilde{f}$ and $\tilde{l}$ will be defined later.

Let $\alpha \in [0,1]$ be a node index in a graphon. Let $[0,T]$ be the time horizon. The value of the a state path, a control path and a Brownian motion at node α at time t are denoted as $x_\alpha(t) \in \mathbb{R}^n$, $u_\alpha(t) \in \mathbb{R}^n$, and $W_\alpha(t) \in \mathbb{R}^n$ respectively.

Let $\mu_{\alpha,t} \in \mathcal{P}_1(\mathbb{R}^n)$ bet the probability measure of the state variable at node α and time t, where $\mathcal{P}_1(\mathbb{R}^n)$ is a probability measure space with the 1-Wasserstein distance. Define the joint probability measure $\nu_{\alpha,t} \in \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^n)$, which is the joint probability measure of the state and control.

Let $\boldsymbol{\mu}_\alpha := \{\mu_{\alpha,t}\}_{0\le t\le T} \in C([0,T], \mathcal{P}_1(\mathbb{R}^n))$ and $\boldsymbol{\nu}_\alpha := \{\nu_{\alpha,t}\}_{0\le t\le T} \in C([0,T], \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^n))$ be the continuous probability measure flow of the state at node α and continuous joint probability measure flow of state and control at node α respectively.

Let $\boldsymbol{\mu}_G := \{\boldsymbol{\mu}_\alpha\}_{\alpha\in[0,1]}$ and $\boldsymbol{\nu}_G := \{\boldsymbol{\nu}_\alpha\}_{\alpha\in[0,1]}$ be the probability measure flow ensemble of the state and joint probability measure flow ensemble of state and control respectively. Let $\boldsymbol{\mu}_G(t)$ and $\boldsymbol{\nu}_G(t)$ be the probability measure ensemble of the state variable at time t, and the joint probability measure ensemble of state and control at time t respectively. We assume that $\boldsymbol{\mu}_G(t) \in C([0,1], \mathcal{P}_1(\mathbb{R}^n))$ and $\boldsymbol{\nu}_G(t) \in C([0,1], \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^n))$ for all $t \in [0,T]$.

Since agents at each node would be governed by a SDE, we need define the probability space. Let $(\Omega, \mathcal{F}, \mathbb{F}_\alpha := \{\mathcal{F}_{\alpha,t}\}_{0\le t\le T}, \mathbb{P}_\alpha)$ be the probability space at node α. And the Brownian motion and state trajectory at node α are adapted to $\mathbb{F}_\alpha$. Assume that that Brownian motion $w_\alpha(t)$ are independent for different α.

B. Problem Setting

Suppose that we are have infinite population on each node α. Define the stochastic control problem that a generic agent at node α faced with,

$$\inf_{\boldsymbol{u}_\alpha \in \mathbb{A}_\alpha(\boldsymbol{\nu}_G)} J_\alpha^{\boldsymbol{\nu}_\alpha}(\boldsymbol{x}_\alpha, \boldsymbol{u}_\alpha) := \mathbb{E}\Big[\int_0^T l(\, x_\alpha^{\boldsymbol{u}_\alpha}(t), u_\alpha(t), \nu_{\alpha,t})dt + g(x_\alpha^{\boldsymbol{u}_\alpha}(T), \mu_{\alpha,T})\Big]$$

$$\text{subject to } \begin{cases} dx_\alpha^{\boldsymbol{u}_\alpha}(t) = f(x_\alpha^{\boldsymbol{u}_\alpha}(t), u_\alpha(t), \boldsymbol{\nu}_G(t))dt \\ \quad +\sigma dw_\alpha(t),\ t \in [0,T], \\ x_\alpha(0) = \xi, \end{cases}$$

where $\boldsymbol{x}_\alpha, \boldsymbol{u}_\alpha$ denote functions on the time horizon $[0,T]$, and l, g, f, σ are functions of appropriate dimensions, and the feasible control space $\mathbb{A}_\alpha$ depend on the joint measure ensemble $\boldsymbol{\nu}_G$ is defined as follows,

$$\mathbb{A}_\alpha(\boldsymbol{\nu}_G) := \Big\{ \boldsymbol{u}_\alpha : \Omega \times [0,T] \to \mathbb{R}^n \ \Big|\ \mathbb{E}\int_0^T |u_\alpha(t)|^2 dt < \infty, \ \mathbb{F}_\alpha\text{−progressively measurable}, \ \boldsymbol{c}_\alpha(\boldsymbol{u}_\alpha, \boldsymbol{\nu}_G) \le \mathbf{0},\ \mathrm{Leb}_1 \otimes \mathbb{P} \text{ a.e.} \Big\} \tag{2}$$

where $\boldsymbol{c}_\alpha : L^2(\Omega \times [0,T]) \times \prod_{\alpha\in[0,1]} C([0,T], \mathcal{P}_1(\mathbb{R}^n)) \ni (\boldsymbol{u}_\alpha, \boldsymbol{\nu}_\alpha) \mapsto \boldsymbol{c}_\alpha(\boldsymbol{u}_\alpha, \boldsymbol{\nu}_\alpha) \in L^\infty(\Omega \times [0,T], \mathbb{R}^{n_c})$ and Leb_1 is the Lebesgue measure on the real line. Let $\boldsymbol{x}_G := \{\boldsymbol{x}_\alpha\}_{\alpha\in[0,1]}$ and $\boldsymbol{u}_G := \{\boldsymbol{u}_\alpha\}_{\alpha\in[0,1]}$ be the trajectory ensemble of state and control respectively.

Degenerate the constraint $\boldsymbol{c}_\alpha$ into the a time wise one c_α defined as $\mathbb{R}^n \times C([0,1], \mathcal{P}_1(\mathbb{R}^n)) \ni (u_\alpha(t), \nu_G(t)) \mapsto c_\alpha(u_\alpha(t), \nu_G(t)) \in \mathbb{R}^{n_c}$. The Hamiltonian of the system at node α and time t would take the following form,

$$H_\alpha(t, x_\alpha(t), u_\alpha(t), \nu_G(t), p_\alpha(t)) := \tilde{l}(x_\alpha(t), u_\alpha(t), \nu_G(t)) + p_\alpha(t) \cdot \tilde{f}(x_\alpha(t), u_\alpha(t), \nu_G(t)) \tag{3}$$

where $p_\alpha(t) \in \mathbb{R}^n$.

Define the following time-wise feasible set of control,

$$A_\alpha(t, \nu_G(t)) := \Big\{ u_\alpha(t)\ \mathbb{R}^n \ \Big|\ c_\alpha(u_\alpha(t), \nu_G(t)) \le 0 \Big\} \tag{4}$$

Define the graphon integral operation on the measure spaces. For any trajectory ensembles $\boldsymbol{x}_G, \boldsymbol{u}_G \in L^2([0,1] \times [0,T]; \mathbb{R}^n)$, any measure ensemble $\boldsymbol{\mu}_G, \boldsymbol{\nu}_G$, and any graphon $\mathcal{W} \in L^2([0,1]^2)$, we define the following network mean fields,

$$\overline{\kappa}_\alpha(t) := \int_0^1 \mathcal{W}(\alpha,\beta) \int_{\mathbb{R}^n} x_\beta(t) \mu_{\beta,t}(dx_\beta(t)) d\beta, \tag{5}$$

$$\overline{\lambda}_\alpha(t) := \int_0^1 \mathcal{W}(\alpha,\beta) \int_{\mathbb{R}^n \times \mathbb{R}^n} (x_\beta(t), u_\beta(t)) \nu_{\beta,t}(dx_\beta(t), du_\beta(t)) d\beta, \tag{6}$$

so the above definitions would give us $\overline{\boldsymbol{\kappa}}_\alpha \in L^2([0,T]; \mathbb{R}^n)$, $\overline{\boldsymbol{\lambda}}_\alpha \in L^2([0,T]; \mathbb{R}^n)$, $\overline{\boldsymbol{\kappa}}_t \in L^2([0,1]; \mathbb{R}^n)$, $\overline{\boldsymbol{\lambda}}_t \in L^2([0,1]; \mathbb{R}^n)$, $\overline{\boldsymbol{\kappa}}_G \in L^2([0,1] \times [0,T]; \mathbb{R}^n)$ and $\overline{\boldsymbol{\lambda}}_G \in L^2([0,1] \times [0,T]; \mathbb{R}^n)$. Except for the expectations of these probability distributions, any moment of them are also of interested, but we will only focus on the expectations in this paper.

Suppose that the constraints $\boldsymbol{c}_\alpha$ only depend on the measure ensemble $\boldsymbol{\nu}_G(t)$ through $\overline{\lambda}_\alpha(t)$, and define the following equivalent constraints depending on $\overline{\lambda}_\alpha(t)$,

$$\psi_\alpha(u_\alpha(t), \overline{\lambda}_\alpha(t)) := c_\alpha(u_\alpha(t), \boldsymbol{\nu}_G(t)). \tag{7}$$

We have the definition of the feasible control set

$$U_\alpha(t, \overline{\lambda}_\alpha(t)) := \Big\{ u_\alpha(t)\ \mathbb{R}^n \ \Big|\ \psi_\alpha(u_\alpha(t), \overline{\lambda}_\alpha(t)) \le 0 \Big\}. \tag{8}$$

We follow a similar setting with [5] by splitting $\tilde{f}, \tilde{l}$ into a local part and an graphon integration part.

Let $\tilde{f}$ and $\tilde{l}$ be defined as follows,

$$\tilde{f}(x, u, \boldsymbol{\nu}_G; g_\alpha) := \int_{\mathbb{R}^n \times \mathbb{R}^n} f_0(x,u,y,v) d\nu_\alpha(y,v) + \int_0^1 \mathcal{W}(\alpha,\beta) \int_{\mathbb{R}^n \times \mathbb{R}^n} f(x,u,y,v) \nu_\beta(dy,dv) d\beta, \tag{9}$$

$$\tilde{l}(x, u, \boldsymbol{\nu}_G; g_\alpha) := \int_{\mathbb{R}^n \times \mathbb{R}^n} l_0(x,u,y,v) d\nu_\alpha(y,v) + \int_0^1 \mathcal{W}(\alpha,\beta) \int_{\mathbb{R}^n \times \mathbb{R}^n} l(x,u,y,v) \nu_\beta(dy,dv) d\beta, \tag{10}$$

where the assumptions on the functions f_0, l_0, f, l will be specified below.

C. Assumptions

Now we consider the following minimized Hamiltonian in a constrained set,

$$\min_{u_\alpha(t)\in U_\alpha(t,\overline{\lambda}_\alpha(t))} H_\alpha(t, x_\alpha(t), u_\alpha(t), \nu_G(t), p_\alpha(t)). \quad (11)$$

In the following paragraph, C is a constant number that may be different at different places.

In order to make the minimized Hamiltonian in this constrained set preserve the properties of local Lipschitz and following linear growth condition required in [8, p.167],

$$|H_\alpha(t, x_\alpha(t), u_\alpha(t), \nu_G(t), p_\alpha(t))| \le C(1 + |p_\alpha(t)|), \quad (12)$$

we make the following assumptions on the functions f_0, f, l_0, l and the sets $U_\alpha(t, \overline{\lambda}_\alpha(t))$ that are similar to the ones in [5], [8].

(H1) For any $\alpha \in [0,1]$, $t \in [0,T]$, $\overline{\lambda}_\alpha(t) \in \mathbb{R}^n$, the set $U_\alpha(t, x_\alpha(t), \overline{\lambda}_\alpha(t)) \subseteq \mathbb{R}^n$ is nonempty, convex and compact. Furthermore, for any $\nu_G(t) \in \mathcal{P}_1(\mathbb{R}^n \times \mathbb{R}^n)$, and any $p_\alpha(t) \in \mathbb{R}^n$, assume that the following argmin set,

$$S_\alpha^{\boldsymbol{\nu}_G(t)}(x_\alpha(t), p_\alpha(t)) \coloneqq \operatorname*{argmin}_{u_\alpha(t)\in U_\alpha(t,\overline{\lambda}_\alpha(t))} H_\alpha(t, x_\alpha(t), u_\alpha(t), \nu_G(t), p_\alpha(t)),$$

is a singleton. And the single solution is Lipschitz in $(x_\alpha(t), p_\alpha(t))$, uniformly on bounded sets.

(H2) There exists a fixed compact set $K \subset \mathbb{R}^n$, such that for any $\alpha \in [0,1]$, any $t \in [0,T]$, and any $\overline{\lambda}_\alpha(t) \in \mathbb{R}^n$, $U_\alpha(t, \overline{\lambda}_\alpha(t)) \subseteq K$.

(H3) For any $\alpha \in [0,1]$, the multifunction $(t, \overline{\lambda}_\alpha(t)) \mapsto U_\alpha(t, \overline{\lambda}_\alpha(t))$ is Lipschitz continuous in the Hausdorff metric, i.e., for any $t, t' \in [0,T]$, any $\overline{\lambda}_\alpha, \overline{\lambda}'_\alpha$,

$$d_H(U_\alpha(t, \overline{\lambda}_\alpha), U_\alpha(t', \overline{\lambda}'_\alpha)) \le C(|t - t'| + |\overline{\lambda}_\alpha - \overline{\lambda}'_\alpha|),$$

where d_H denotes the Hausdorff distance between sets.

(H4) The functions f_0, f, l_0, l are bounded and continuous functions on $\mathbb{R}^n \times K \times \mathbb{R}^n \times K$. And Lipshitz continuous in (x, y) uniformly with respect to (u, v);

(H5) The functions f_0, f are Lipschitz continuos in (u, v), uniformly with respect to (x, y).

(H6) For any bounded and measurable function $\zeta(\beta)$, the mapping $\alpha \mapsto \int_0^1 g(\alpha, \beta)\zeta(\beta)d\beta$ is continuous in α.

(H7) For the graphon $\mathcal{W}(\alpha, \beta)$, we assume that for all $\alpha \in [0,1]$,

$$\sup_{\alpha\in[0,1]} \int_0^1 |\mathcal{W}(\alpha, \beta)| d\beta < \infty. \quad (13)$$

In the subsequent paragraph, $\boldsymbol{\nu}_G$ may also be written as $\nu_G(\cdot)$ as they have the same meaning.

We have to restrict the start point of the above loop, the joint probability flow ensemble $\nu_G(\cdot)$ into the space $\mathcal{V}_{[0,T]}$, following the similar idea in [5].

(C1) For any $\alpha \in [0,1]$ and $t \in [0,T]$, $\nu_G(t) \in \mathcal{P}_1([0,1], \mathbb{R}^n \times \mathbb{R}^n)$.

(C2) There exists a $\eta \in (0,1]$, such that for any bounded and Lipschitz continuous function $\phi(x,u)$ on $\mathbb{R}^n$,

$$\sup_{\beta\in[0,1]} \Big| \int_{\mathbb{R}^n\times\mathbb{R}^n} \phi(x,u)\nu_\beta(t_1, dx, du) - \int_{\mathbb{R}^n\times\mathbb{R}^n} \phi(x,u)\nu_\beta(t_2, dx, du)\Big| \le C_h|t_1 - t_2|^\eta, \quad (14)$$

where the constant C_h may depend on the Lipschitz constant $Lip(\phi)$.

We have another definition of the measure ensembles built on the space of continuous function $C_T \coloneqq C([0,T], \mathbb{R}^n \times \mathbb{R}^n)$. We use $\mathcal{F}_T$ to denote the σ-algebra generate by the cylindrical sets of the form $\{x(\cdot) \in C_T : x(t_i) \in B_i,\ 1 \le i \le j \text{ for some } j\}$, where B_i is a Borel set. On the space C_T, we equip the metric $\rho(x,y) = \sup_t |x(t) - y(t)| \wedge 1$. Thus the metric space (C_T, ρ) is complete. Let $\mathbf{V}_{\alpha,T}$ be the probability measure space on $(C_T, \mathcal{F}_T)$ at node α. For an $\omega \in C_T$, define the canonical process $X_t(\omega) \coloneqq \omega(t)$. For any two element $\nu_1, \nu_2 \in \mathbf{V}_{\alpha,T}$, we define the following Wasserstein distance between them depending upon the original metric ρ in the space C_T,

$$D_T(\nu_1, \nu_2) \coloneqq \inf_{\widehat{\pi}} \int_{C_T\times C_T} \left(\sup_{s\le T} |X_s(\omega_1) - X_s(\omega_2) \wedge 1|\right) d\widehat{\pi}(\omega_1, \omega_2), \quad (15)$$

where $\widehat{\pi}$ is a probability measure on the $(C_T, \mathcal{F}_T)\times(C_T, \mathcal{F}_T)$ with marginals ν_1 and ν_2. Then $(\mathbf{V}_{\alpha,T}, D_T)$ is a complete metric space [9]. Let $\mathbf{V}_T^G \coloneqq \prod_{\alpha\in[0,1]} \mathbf{V}_{\alpha,T}$.

Define a metric on this joint measure ensemble space $\mathbf{V}_T^G$. Let $\nu_G \coloneqq \{\nu_\alpha\}_{\alpha\in[0,1]}$ and $\tilde{\nu}_G \coloneqq \{\tilde{\nu}_\alpha\}_{\alpha\in[0,1]}$ be two joint measure ensembles, define the metric as $d(\nu_G, \tilde{\nu}_G) \coloneqq \sup_{\alpha\in[0,2]} D_T(\nu_G, \tilde{\nu}_G)$. And we have that the joint measure space under this metric is complete, which will be shown in Lemma 1.

III. ANALYSIS ON THE HJB EQUATION WITH A GIVEN PROBABILITY MEASURE FLOW

Lemma 1: The space $(\mathbf{V}_T^G, d)$ is a complete metric space.

Proof: For any Cauchy sequence $\{\nu_G^k, k \ge 1\}$ in the space $\mathbf{V}_T^G$, its projection $\{\mathrm{Proj}_\alpha(\nu_G^k), k \ge 1\}$ would also be a Cauchy sequence in the complete metric spaces $\mathbf{V}_{\alpha,T}$. Thus those projections admit a limit in each $\mathbf{V}_{\alpha,T}$. The element comprises of those limits give a limit of the original Cauchy sequence. ■

We will construct a contraction mapping on the above complete metric space. However, to facilitate the analysis on the HJB equation, we start from a fixed joint probability flow ensemble $\nu_G(\cdot) \in \mathcal{V}_{[0,T]}$, which will yields a classical solution to the HJB as it will be shown in Lemma 4.

Note that we can always select such a probability measure flow ensemble despite there are constraints on the control, which would make its distribution at some time instant not absolutely continuous, but the probability measure always exits. And we use the following notations for the fixed dynamics, running costs, and network control mean fields,

$$\tilde{f}^*_\alpha(t,x,u) := \tilde{f}(x,u,\nu_G;g_\alpha),\ \tilde{l}^*_\alpha(t,x,u) := \tilde{f}(x,u,\nu_G;g_\alpha),$$
$$\overline{\lambda}^*_\alpha(t) := \int_0^1 \mathcal{W}(\alpha,\beta)\int_{\mathbb{R}^n\times\mathbb{R}^n}(x_\beta(t),u_\beta(t)) \nu_{\beta,t}(dx_\beta(t),du_\beta(t))d\beta. \tag{16}$$

Now in this fixed $\nu_G(\cdot)$, the HJB would read as,

$$-V^\alpha_t(t,x) = \inf_{u\in U_\alpha(t,\overline{\lambda}^*_\alpha)}(\tilde{f}^*_\alpha(t,x,u)\cdot V^\alpha_x(t,x)+\tilde{l}^*_\alpha(t,x,u)) + \frac{\sigma^2}{2}V^\alpha_{xx}(t,x),\ V^\alpha(T,x)=g(x). \tag{17}$$

In this particular case where $\nu_G(\cdot)$ is assumed to be fixed, we define the minimized Hamiltonian as follows,

$$\widehat{H}_\alpha(t,x,p) := \min_{u\in U_\alpha(t,\overline{\lambda}^*_\alpha)}\tilde{f}^*_\alpha(t,x,u)\cdot p+\tilde{l}^*_\alpha(t,x,u). \tag{18}$$

We have the following Lipshitz continuity and Hölder continuity properties on the dynamics $\tilde{f}^*_\alpha(t,x,u)$ and cost function $\tilde{l}^*_\alpha(t,x,u)$.

Lemma 2: For the functions $h_\alpha = f^*_\alpha(t,x,u)$ or $l^*_\alpha(t,x,u)$, there exist positive constants C_x, C_u, $C_t>0$ such that h_α is Lipschitz continuous in x and u with Lipshitz constants C_x and C_u, and Hölder continuous in t with constant C_t and exponent η.

Proof: We demonstrate the proof of $\tilde{f}^*_\alpha$, and the proof for $\tilde{l}^*_\alpha$ is identical.

(i) Lipschitz continuity in x and u:

By assumptions (H4) and (H5), the base functions f_0 and f are Lipschitz continuous in (x,u) uniformly with respect to (y,v). Let L_f be the maximum of their Lipschitz constants. For any $x_1,x_2\in\mathbb{R}^n$, we have,

$$\begin{aligned}&|\tilde{f}^*_\alpha(t,x_1,u)-\tilde{f}^*_\alpha(t,x_2,u)|\\ &\le \int_{\mathbb{R}^{2n}}|f_0(x_1,\cdot)-f_0(x_2,\cdot)|d\nu_{\alpha,t}\\ &\quad+\int_0^1\mathcal{W}(\alpha,\beta)\int_{\mathbb{R}^{2n}}|f(x_1,\cdot)-f(x_2,\cdot)|d\nu_{\beta,t}d\beta\\ &\quad L_f|x_1-x_2|\nu_{\alpha,t}(\mathbb{R}^{2n})+\left(\sup_\alpha\int_0^1\mathcal{W}_{\alpha,\beta}d\beta\right)\\ &\quad L_f|x_1-x_2|\nu_{\beta,t}(\mathbb{R}^{2n}).\end{aligned} \tag{19}$$

Since the probability measures integrate to 1, and by (H7) the graphon integral is bounded by some constants $M_{\mathcal{W}}$, we obtain $|\tilde{f}^*_\alpha(t,x_1,u)-\tilde{f}^*_\alpha(t,x_2,u)|\le L_f(1+M_{\mathcal{W}})|x_1-x_2|$. The exact same argument applies to u.

(ii) Hölder continuity in t:

By the definition of the fixed environment space (C2), the probability measure flow $t\mapsto\nu_G(t)$ is Hölder continuous under Wasserstein metric W_1 with exponent η. Since f_0 and f are Lipschitz continuous in (y,v), we can use the Kantorovich- Rubinstein duality for the Wasserstein distance. For any $t,s\in[0,T]$,

$$\begin{aligned}|\tilde{f}^*_\alpha(t,x,u)-\tilde{f}^*_\alpha(s,x,u)| &\le \mathrm{Lip}(f_0)W_1(\nu_{\alpha,t},\nu_{\alpha,s})\\ &+M_{\mathcal{W}}\sup_\beta(\mathrm{Lip}(f)W_1(\nu_{\beta,t},\nu_{\beta,s})).\end{aligned} \tag{20}$$

By the Hölder property of the measure flow, $W_1(\nu_{\beta,t},\nu_{\beta,s})\le C_h|t-s|^\eta$. Therefore,

$$|\tilde{f}^*_\alpha(t,x_1,u)-\tilde{f}^*_\alpha(t,x_2,u)|\le C|t-s|^\eta, \tag{21}$$

completing the proof. ■

We need some regularity properties of the minimized Hamiltonian $\widehat{H}_\alpha(t,x,p)$ in order to apply the linear PDE theory from [10].

Lemma 3: For any $\alpha\in[0,1]$, any x,y,p,q in a compact subset of $\mathbb{R}^n$, and any $t,s\in[0,T]$, there exist constants $C,\eta>0$ such that the following inequalities are satisfied,

$$\begin{aligned}|\widehat{H}_\alpha(t,x,p)-\widehat{H}_\alpha(s,x,p)| &\le C(1+|p|)|t-s|^\eta,\\ |\widehat{H}_\alpha(t,x,p)-\widehat{H}_\alpha(t,y,p)| &\le C(1+|p|)|x-y|,\\ |\widehat{H}_\alpha(t,x,q)-\widehat{H}_\alpha(t,x,q)| &\le C|p-q|.\end{aligned} \tag{22}$$

Moreover, $\widehat{H}_\alpha(t,x,p)$ is continuous in α.

Proof: Let the pre-minimized function be denoted as $J_\alpha := \tilde{f}^*_\alpha(t,x,u)\cdot p+\tilde{l}^*_\alpha(t,x,u)$. By Lemma 2, J_α is Lipschitz in x,u,p and Hölder continuous in t. By (H2), the control u is constrained in a uniformly compact set K.

(i) Lipschitz continuity in p:

$$\begin{aligned}&|\widehat{H}_\alpha(t,x,p)-\widehat{H}_\alpha(t,x,q)|\\ &=\Big|\min_{u\in U_\alpha}J_\alpha(t,x,u,p)-\min_{u\in U_\alpha}J_\alpha(t,x,u,q)\Big|\\ &\le\max_{u\in U_\alpha}|J_\alpha(t,x,u,p)-J_\alpha(t,x,u,q)|\\ &\le\max_{u\in K}|\tilde{f}^*_\alpha(t,x,u)\cdot(p-q)|\\ &\le\|\tilde{f}^*_\alpha\|_\infty|p-q|\le C|p-q|,\end{aligned} \tag{23}$$

where the boundedness of $\tilde{f}^*_\alpha$ follows from (H4).

(ii) Lipschitz continuity in x:

Let u_x and u_y be the minimizer of $\tilde{H}_\alpha$ evaluated at x and y, respectively. Then,

$$\begin{aligned}&\widehat{H}_\alpha(t,x,p)-\widehat{H}_\alpha(t,y,p)\le J_\alpha(t,x,u_y,p)-J_\alpha(t,y,u_y,p)\\ &=(\tilde{f}^*_\alpha(t,x,u_y)-\tilde{f}^*_\alpha(t,y,u_y))\cdot p\\ &\quad+(\tilde{l}^*_\alpha(t,x,u_y)-\tilde{l}^*_\alpha(t,y,u_y)).\end{aligned} \tag{24}$$

By the Lipschitz continuity of $\tilde{f}^*_\alpha$ and $\tilde{l}^*_\alpha$ in x established in Lemma 2 (with constant C_x),

$$\begin{aligned}&\widehat{H}_\alpha(t,x,p)-\widehat{H}_\alpha(t,y,p)\le C_x|x-y||p|+C_x|x-y|\\ &C_x(1+|p|)|x-y|.\end{aligned} \tag{25}$$

Reversing the roles of x and y yields the symmetric lower bound, establishing $|\widehat{H}_\alpha(t,x,p)-\widehat{H}_\alpha(t,y,p)|\le C(1+|p|)|x-y|$.

(iii) Hölder continuity in t:

This steps requires for moving constraints. Let $u_s\in U_\alpha(s,\overline{\lambda}_\alpha(s))$ be the minimizer at time s. By (H3), the constraint set is Lipschitz in the Hausdorff metric d_H with respect to $(t,\overline{\lambda}_\alpha)$. Since $\nu_G(\cdot)$ is Hölder continuous in time,

$\overline{\lambda}_\alpha(t)$ is also η-Hölder in time. Thus,

$$d_H(U_\alpha(t,\overline{\lambda}_\alpha(t)), U_\alpha(s,\overline{\lambda}_\alpha(s))) \leq C(|t-s| + |\overline{\lambda}_\alpha(t) - \overline{\lambda}_\alpha(s)|) \leq C_1|t-s|^\eta. \quad (26)$$

This implies that there exists a feasible control at time t, $u_t \in U_\alpha(t,\overline{\lambda}_\alpha(t))$, such that $|u_t - u_s| \leq C_1|t-s|^\eta$.

We bound the Hamiltonian difference by evaluating the t-Hamiltonian at this suboptimal u_t,

$$\widehat{H}_\alpha(t,x,p) - \widehat{H}_\alpha(s,x,p) \leq J_\alpha(t,x,u_t,p) - J_\alpha(s,x,u_s,p).$$

Add and subtract $J_\alpha(s,x,u_t,p)$ on the above s expression,

$$[J_\alpha(t,x,u_t,p) - J_\alpha(s,x,u_t,p)] + [J_\alpha(s,x,u_t,p) - J_\alpha(s,x,u_s,p)]. \quad (27)$$

The first bracket is bounded by $C_2(1+|p|)|t-s|^\eta$ due to the Hölder continuity of $\tilde{f}^*_\alpha$ and $\tilde{l}^*_\alpha$ in time (Lemma 2). The second bracket is bounded by $C_u(1+|p|)(u_t - u_s)$ due to the Lipschitz continuity in u (Lemma 2).
Substituting the Hausdorff bound $|u_t - u_s| \leq C_1|t-s|^\eta$,

$$\widehat{H}_\alpha(t,x,p) - \widehat{H}_\alpha(s,x,p) \leq C_2(1+|p|)|t-s|^\eta + C_uC_1(1+|p|)|t-s|^\eta \leq C(1+|p|)|t-s|^\eta. \quad (28)$$

Symmetry yields the absolute value.

(iv) Continuity in α:

To establish continuity in α, we note that the inner integral of the network control mean field $\zeta(\beta) := \int u d\nu_{\beta,t}$ is a bounded measurable function due to the compactness of the control space (H2). By the property of the graphon assumed in (H6), the mapping $\alpha \mapsto \overline{\lambda}_\alpha(t) = \int_0^1 \mathcal{W}(\alpha,\beta)\zeta(\beta)d\beta$ is continuous. Since the constraint set U_α is continuous with respect to $\overline{\lambda}_\alpha$ in the Hausdorff metric (H3), and the basis functions are uniformly continuous, the minimized Hamiltonian $\widehat{H}_\alpha$ inherits continuity in α. ■

Following is a existence result about the above HJB (17) similar to [5], due to we have already obtained the Lipschitz continuity linear growth conditions of the Hamiltonian.

Lemma 4 (Solvability of HJB): Let assumptions (H1)-(H7) hold. For any fixed joint measure flow $\nu_G \in \mathcal{V}_{[0,T]}$, the resulting terminal value problem for the HJB equation (17) admits a unique classical solution $V^\alpha \in C_b^{2,1}([0,T]\times\mathbb{R}^n)$. Furthermore, there exists a constant C_V, independent of α and ν_G, such that the spatial gradient is uniformly bounded and globally Lipschitz in x,

$$\sup_{\alpha,t,x}|V_x^\alpha(t,x)| \leq C_V, \quad \sup_{\alpha,t,x\neq y}\frac{|V_x^\alpha(t,x) - V_x^\alpha(t,y)|}{|x-y|} \leq C_V.$$

Proof: By assumptions (H1), (H2), and (H4), the constrained control set $U_\alpha(t,\overline{\lambda}_\alpha(t))$ is uniformly compact, and the functions $\tilde{f}$ and $\tilde{l}$ are uniformly bounded. Consequently, the minimized Hamiltonian $\widehat{H}_\alpha(t,x,p) := \min_{u\in U_\alpha}\{\tilde{f}^*_\alpha(t,x,u)\cdot p + \tilde{l}^*_\alpha(t,x,u)\}$ satisfies a linear growth condition in the momentum variable p, i.e., $|\widehat{H}_\alpha(t,x,p)| \leq C(1+|p|)$.

Furthermore, by Lemma 3, $\widehat{H}_\alpha$ is locally Hölder continuous in t and Lipschitz continuous in x and p. Because the diffusion coefficient $\sigma^2/2$ is strictly positive and constant, the PDE is uniformly parabolic.

Applying the classical existence and regularity theory for semilinear parabolic equations (specifically, the maximum principle and gradient estimates established in [10]), there exists a unique classical solution $V^\alpha \in C_b^{2,1}$. The uniform bounds on the Hölder/Lipschitz estimates on V_x^α follow directly from the global bounds on the coefficients and linear growth of the Hamiltonian, independent of the specific index α or the fixed measure flow ν_G. ■

Corollary 5 (Closed-Loop Well-Posedness): For any fixed $\nu_G \in \mathcal{V}_{[0,T]}$, the optimal feedback law is given by,

$$\phi_\alpha(t,x|\nu_G) := \arg\min_{u\in U_\alpha(t,\overline{\lambda}_\alpha(t))}\Big\{\tilde{f}(x,u,\nu_G)\cdot V_x^\alpha(t,x) + \tilde{l}(x,u,\nu_G)\Big\} \quad (29)$$

is uniformly bounded and globally Lipschitz continuous with respect to x. Consequently, the following closed-loop SDE,

$$dx_\alpha(t) = \tilde{f}(x_\alpha(t), \phi(t,x_\alpha(t)|\nu_G(t)), \nu_G(t); g_\alpha)dt + \sigma dw_\alpha(t) \quad (30)$$

satisfies the standard Itô conditions and admits a unique strong solution.

Proof: The uniform boundedness of ϕ_α is guaranteed by the compactness of the constraint set K (H2). By (H1), the $\arg\min$ is a singleton. Since $\tilde{f}$ and $\tilde{l}$ are Lipschitz in x and u, and V_x^α is Lipschitz in x (from Lemma 4), the strict convexity of the Hamiltonian guarantees that the unique minimizer ϕ_α inherits the Lipschitz continuity in x. Thus Lipschitz property of the drift ensures of a unique strong solution to the associated SDE. ■

IV. EXISTENCE OF THE CONSISTENT JOINT MEASURE FLOW

In this section, we establish the existence of a fixed point in the space of joint probability measure flows. The primary challenge in the Extended GMFG framework is that the control constraints depend dynamically on the network control mean field, $\overline{\lambda}_\alpha(t)$, which in turn is governed by the joint measure flow ν_G.

We begin by defining the appropriate metric space for the adjoint measure flow and the pushforward mechanism induced by the optimal feedback laws.

Definition 6 (Joint Measure Disintegration): Recall that $\mathcal{V}_{[0,T]}$ is the space of the joint probability measure flows. Equip it with the metric $D_T(\nu_G,\nu'_G) := \sup_{\alpha\in[0,1]}\sup_{t\in[0,T]}W_1(\nu_{\alpha,t},\nu'_{\alpha,t})$, where W_1 is the 1-Wasserstein metric. For any fixed $\nu_G \in \mathcal{V}_{[0,T]}$, let $\phi_\alpha(t,x|\nu_G)$ be the unique optimal feedback law derived from (29). If $\mu_{\alpha,t}$ is the marginal state measure, the consistent joint measure $\nu^{new}_{\alpha,t}$ is defined by the disintegration,

$$\int_{\mathbb{R}^n\times\mathbb{R}^n}\psi(x,u)\nu^{new}_{\alpha,t}(dx,du) = \int_{\mathbb{R}^n}\psi(x,\phi_\alpha(t,x|\nu_G))\mu_{\alpha,t}(dx) \quad (31)$$

for any bounded continuous test function ψ. We denote this concisely as $\nu^{new}_{\alpha,t} = \mu_{\alpha,t} \otimes \delta_{\phi_\alpha(t,x|\nu_G)}$.

To establish a contraction mapping, we first must prove that the optimal feedback law is sufficiently regular (Lipschitz continuous) with respect to the underlying measure flow. This is the analogue of the sensitivity condition in standard MFG, extended to the graphon and constrained control setting.

Lemma 7 (Sensitivity of the Best Response): Under assumptions (H1)-(H7), there exists a constant C_ϕ such that for any two joint measure flows $\nu_G, \nu'_G \in \mathcal{V}_{[0,T]}$, the corresponding optimal feedback laws satisfy,

$$\begin{aligned}\sup_{\alpha\in[0,1]} \sup_{t\in[0,T],x\in\mathbb{R}^n} &|\phi_\alpha(t,x|\nu_G) - \phi_\alpha(t,x|\nu'_G)| \\ &\leq C_\phi D_T(\nu_G,\nu'_G). \end{aligned} \tag{32}$$

Proof: By (H1), the Hamiltonian H_α is strictly convex with respect to u, yielding a unique minimizer. By (H3), the feasible control set $U_\alpha(t,\overline{\lambda}_\alpha(t))$ is Lipschitz continuous in the Hausdorff metric d_H with respect to $\overline{\lambda}_\alpha(t)$. By standard parametric optimization results for strictly convex functions on compact, Hausdorff-Lipschitz moving domains, the minimizer is Lipschitz continuous with respect to the parameter $\overline{\lambda}_\alpha(t)$. Therefore, there exits C_3 such that,

$$|\phi_\alpha(t,x|\nu_G) - \phi_\alpha(t,x|\nu'_G)| \leq C_3|\overline{\lambda}_\alpha(t) - \overline{\lambda}'_\alpha(t)|. \tag{33}$$

Recall the definition of the joint network control mean field from (6),

$$\overline{\lambda}_\alpha(t) = \int_0^1 \mathcal{W}(\alpha,\beta)\int_{\mathbb{R}^n\times\mathbb{R}^n}(x,u)\nu_{\beta,t}(dx,du)d\beta. \tag{34}$$

Let $\iota(x,u) := (x,u)$ be the identity mapping on $\mathbb{R}^n\times\mathbb{R}^n$. Under the standard Euclidean norm, ι is globally Lipschitz continuous with a Lipschitz constant of $L_\iota = 1$. By the Kantorovich- Rubinstein duality theorem for the 1-Wasserstein metric, the difference in the inner integrals can be bounded directly by the Wasserstein distance between the two joint measures,

$$\begin{aligned}\Big|\int_{\mathbb{R}^{2n}}(x,u)\nu_{\beta,t}(dx,du) - \int_{\mathbb{R}^{2n}}(x,u)\nu'_{\beta,t}(dx,du)\Big| \\ \leq W_1(\nu_{\beta,t}\nu'_{\beta,t}). \end{aligned} \tag{35}$$

Applying this bound to the definition of the network mean field yields,

$$\begin{aligned}|\overline{\lambda}_\alpha(t) - \overline{\lambda}'_\alpha(t)| &\leq \int_0^1 \mathcal{W}(\alpha,\beta)W_1(\nu_{\beta,t},\nu'_{\beta,t})d\beta \\ &\leq \left(\sup_{\alpha\in[0,1]}\int_0^1\mathcal{W}(\alpha,\beta)d\beta\right)\cdot\sup_{\beta\in[0,1]}W_1(\nu_{\beta,t},\nu'_{\beta,t}). \end{aligned} \tag{36}$$

By assumption (H7), the graphon integral is bounded by a finite constant $M_\mathcal{W}$. Substituting the definition of the joint measure flow metric $D_T(\nu_G,\nu'_G)$, we obtain,

$$|\overline{\lambda}_\alpha(t) - \overline{\lambda}'_\alpha(t)| \leq M_\mathcal{W} D_T(\nu_G,\nu'_G). \tag{37}$$

Combining the inequalities yields the desired sensitivity results,

$$|\phi_\alpha(t,x|\nu_G) - \phi_\alpha(t,x|\nu'_G)| \leq C_3 M_\mathcal{W} D_T(\nu_G,\nu'_G) \tag{38}$$

which holds with $C_\phi = C_3 M_\mathcal{W}$. ∎

We now define the operator mapping an assumed environment ν_G to the realized environment. To ensure the resulting probability measure flow satisfies the continuity properties of $\mathcal{V}_{[0,T]}$, we must first identify a consistent trajectory ensemble in the path space $\mathbf{V}^G_T$.

Before defining the mapping on the path space, we explicitly establish the marginal projection operator. Given a probability measure $\mathbf{P}_\alpha \in \mathbf{V}_{\alpha,T}$, define the time marginal $\nu_\alpha(t)$ by

$$\nu_\alpha(t,B) := \mathbf{P}_\alpha((x_\alpha,u_\alpha)\in C_T : (x_\alpha,x_\alpha)\in B), \tag{39}$$

for any Borel set $B\subset\mathbb{R}^{2n}$. We denote this mapping from the path measure space $\mathbf{V}_{\alpha,T}$ to the state probability measure space $\mathcal{P}_1(\mathbb{R}^n)$ as,

$$\mathrm{Marg}_t(\mathbf{P}_\alpha) = \nu_\alpha(t). \tag{40}$$

Consider a path measure ensemble $\mathbf{P}\in\mathbf{V}^G_T$. Define the time marginal mapping for the entire ensemble by,

$$\mathrm{Marg}_t(\mathbf{P}) = (\mathrm{Marg}_t(\mathbf{P}_\alpha))_{\alpha\in[0,1]}. \tag{41}$$

Definition 8 (Generalized MV Mapping on Path Space): For any fixed assumed joint measure flow $\nu_G\in\mathcal{V}_{[0,T]}$, let $\phi_\alpha(t,x|\nu_G)$ be the unique optimal feedback law derived from (29). We define a mapping $\Phi(\cdot|\nu_G) : \mathbf{V}^G_T \to \mathbf{V}^G_T$ as follows. For any test path measure ensemble $\mathbf{P}_G\in\mathbf{V}^G_T$, we construct the induced joint measure flow,

$$\overline{\nu}_{\alpha,t} := \mathrm{Marg}_t(\mathbf{P}_\alpha)\otimes\delta_{\phi_\alpha(t,\cdot|\nu_G)}. \tag{42}$$

And solve the following closed-loop SDE for all $\alpha\in[0,1]$,

$$\begin{aligned}dx_\alpha(t) = \tilde{f}(x_\alpha(t),\phi_\alpha(t,x_\alpha(t)|\nu_G),\overline{\nu}_G[\mathbf{P}_G](t);g_\alpha)dt \\ +\sigma dw_\alpha(t), \end{aligned} \tag{43}$$

with $x_\alpha(0)\sim\mu^\alpha_0$. The output $\mathbf{P}^{new}_G = \Phi_\mathbf{V}[\mathbf{P}_G|\nu_G]$ is defined as the path law ensemble of the solutions $x_\alpha(\cdot)$.

Lemma 9 (Uniqueness of the Consistent Path Element): Under assumptions (H1)-(H7), for any fixed $\nu_G\in\mathcal{V}_{[0,T]}$, the mapping $\Phi_\mathbf{V}[\cdot|\nu_G]$ admits a unique fixed point $\mathbf{P}^*_G\in\mathbf{V}^G_T$. We define $\mathbf{V}^{G1}_T\subset\mathbf{V}^G_T$ as the set of all such unique fixed points generated by all $\nu_G\in\mathcal{V}_{[0,T]}$. Furthermore, the marginal flow of any element in $\mathbf{V}^{G1}_T$ belongs to $\mathcal{V}_{[0,T]}$.

Proof: Let $\mathbf{P}_G,\mathbf{P}'_G\in\mathbf{V}^G_T$. We evaluate the 1-Wasserstein distance between their induced joint marginals $\overline{\nu}_{\alpha,t}$ and $\overline{\nu}'_{\alpha,t}$. For any 1-Lipschitz in test function $\psi(x,u)$, let $\tilde{\psi}(x) := \psi(x,\phi_\alpha(t,x|\nu_G))$. Since Φ_α is L_ϕ-Lipschitz in x (from Corollary 5), $\tilde{\psi}$ is $(1+L_\phi)$-Lipschitz. By Kantorovich-Rubinstein duality,

$$\begin{aligned}&W_1(\overline{\nu}_{\alpha,t}[\mathbf{P}_G],\overline{\nu}'_{\alpha,t}[\mathbf{P}'_G]) \\ &\quad\leq(1+L_\phi)W_1(\mathrm{Marg}_t(\mathbf{P}_\alpha),\mathrm{Marg}_t(\mathbf{P}'_\alpha)) \\ &\quad\leq(1+L_\phi)D_t(\mathbf{P}_\alpha,\mathbf{P}'_\alpha). \end{aligned} \tag{44}$$

Let x_α, x'_α be the strong solutions to (43) driven by $\mathbf{P}_G$ and $\mathbf{P}'_G$ under the same Brownian motion. By the Lipschitz continuity of $\tilde{f}$,

$$|x_\alpha(t) - x'_\alpha(t)| \le L_f \int_0^t \Big(|x_\alpha(s) - x'_\alpha(s)| + \sup_\beta W_1(\overline{\nu}_{\beta,s}[\mathbf{P}_G], \overline{\nu}'_{\beta,s}[\mathbf{P}'_G]) \Big) ds. \tag{45}$$

Substituting (44) and taking the supremum over α and expectations yields,

$$D_t(\Phi_{\mathbf{V}}[\mathbf{P}_G], \Phi_{\mathbf{V}}[\mathbf{G}']) \le L_f \int_0^t \Big(D_s(\Phi_{\mathbf{V}}[\mathbf{P}_G], \Phi_{\mathbf{V}}[\mathbf{P}'_G]) + (1 + L_\phi) D_s(\mathbf{P}_G, \mathbf{P}'_G) \Big) ds. \tag{46}$$

Applying Gronwall's lemma and standard Volterra iteration establishes that $\Phi_{\mathbf{V}}^{k_0}$ is a strict contraction for a sufficient large k_0. Thus, a unique fixed point $\mathbf{P}^*_G \in \mathbf{V}^G_T$ exists. Since the SDE (43) has a bounded drift and constant diffusion, standard moment estimates guarantee the marginal flow $\mathrm{Marg}_t(\mathbf{P}^*_\alpha)$ is $1/2$-Hölder continuous in time, satisfying condition (C2). Therefore, the consistent marginal flow $\nu_G^{new} := \overline{\nu}_G[\mathbf{P}^*_G] \in \mathcal{V}_{[0,T]}$. ■

We now evaluate the sensitivity of these fixed points in $\mathbf{V}^{G1}_T$ with respect to the original assumed flows.

Lemma 10 (Sensitivity of Consistent Path Measures): Let $\nu_G, \nu'_G \in \mathcal{V}_{[0,T]}$ be two assumed flows, generating best responses $\phi_\alpha, \phi'_\alpha$. Let $\mathbf{P}_G, \mathbf{P}'_G \in \mathbf{V}^{G1}_T$ be their corresponding unique fixed-point path ensembles form Lemma 9. There exists a constant $C_{\mathbf{V}}$ such that,

$$D_T(\mathbf{P}_G, \mathbf{P}'_G) \le C_{\mathbf{V}} \sup_{\alpha,t,x} |\phi_\alpha(t, x|\nu_G) - \phi'_\alpha(t, x|\nu_G)|.$$

Proof: For $\mathbf{P}_G$ and $\mathbf{P}'_G$, the respective induced marginal flows are $\overline{\nu}_{\alpha,t} = \nu_{\alpha,t} \otimes \delta_{\phi_\alpha}$ and $\overline{\nu}'_{\alpha,t} = \mu'_{\alpha,t} \otimes \delta_{\phi'_\alpha}$. Similar to (44), evaluating the Wasserstein distance with the added policy difference gives,

$$W_1(\overline{\nu}_{\alpha,t}, \overline{\nu}'_{\alpha,t}) \le (1 + L_\phi) D_t(\mathbf{P}_\alpha, \mathbf{P}'_\alpha) + \sup_x |\phi_\alpha(t, x|\nu_G) - \phi'_\alpha(t, x|\nu'_G)|. \tag{47}$$

Let x_α, x'_α be the strong solutions of their respective SEDs (43). Taking difference of the state trajectories,

$$|x_\alpha(t) - x'_\alpha(t)| \le L_f \int_0^t \Big(|x_\alpha(s) - x_\alpha(s)'| + |\phi_\alpha - \phi'_\alpha + \sup_\beta W_1(\overline{\nu}_{\beta,s}, \overline{\nu}'_{\beta,s})| \Big) ds. \tag{48}$$

Substituting (47) into the integral and defining $\Delta_\phi := \sup_{\alpha,t,x}(\phi_\alpha - \phi'_\alpha)$,

$$D_t(\mathbf{P}_G, \mathbf{P}'_G) \le L_f \int_0^t \Big((1 + (1 + L_\phi)) D_s(\mathbf{P}_G, \mathbf{P}'_G) + 2\Delta_\phi \Big) ds. \tag{49}$$

Applying Gronwall's lemma directly yields $D_T(\mathbf{P}_G, \mathbf{P}'_G) \le C_{\mathbf{V}} \Delta_\phi$, where $C_{\mathbf{V}} = 2L_f T \exp(L_f(2 + L_\phi)T)$. ■

We now define the contraction mapping on the joint measure flow space.

Definition 11 (The Fixed-Point Mapping on Path Space): Define the mapping $\Psi : \mathbf{V}^G_T \to \mathbf{V}^G_T$. For any path measure ensemble $\mathbf{P}_G \in \mathbf{V}^G_T$, let $\nu_G = \mathrm{Marg}(\mathbf{P}_G)$ be its induced joint marginal flow. Define $\Psi(\mathbf{P}_G) := \mathbf{P}^*_G$, where $\mathbf{P}^*_G$ is the uniques consistent path measure ensemble generated under the assumed environment ν_G, as established in Lemma 9.

(H8) The model parameter satisfy the weak coupling condition $C_\phi C_{\mathbf{V}} < 1$, where C_ϕ and $C_{\mathbf{V}}$ are the constants defined in Lemma 7 and Lemma 10, respectively.

Theorem 12 (Existence of the Nash Equilibrium): Under assumptions (H1)-(H8), the mapping Ψ admits a unique fixed point $\mathbf{P}^*_G \in \mathbf{V}^G_T$. The induced flow $\nu^*_G = \mathrm{Marg}(\mathbf{P}^*_G)$, together with the feedback strategy $\phi_\alpha(t, x|\nu^*_G)$, constitutes a unique Nash equilibrium for the EGMFG.

Proof: Let $\mathbf{P}_G, \mathbf{P}'_G \in \mathbf{V}^G_T$, and denote their corresponding joint marginal flows by $\nu_G = \mathrm{Marg}(\mathbf{P}_G)$ and $\tilde{\nu}_G = \mathrm{Marg}(\tilde{\mathbf{P}}_G)$. By Lemma 10, the distance between the mapped path ensembles is bounded by the difference in their induced feedback policies,

$$D_T(\Psi(\mathbf{P}_G), \Psi(\tilde{\mathbf{P}}_G)) \le C_{\mathbf{V}} \sup_{\alpha,t,x} |\phi_\alpha(t, x|\nu_G) - \tilde{\phi}_\alpha(t, x|\tilde{\nu}_G)|. \tag{50}$$

Applying the sensitivity bound of the best response from Lemma from Lemma 7 gives,

$$D_T(\Psi(\mathbf{P}_G), \Psi(\tilde{\mathbf{P}}_G)) \le C_\phi C_{\mathbf{V}} D_T(\nu_G, \tilde{\nu}_G). \tag{51}$$

By definition, the Wasserstein distance between state marginals is bounded by the supremum norm distance over the trajectory space, which implies $D_T(\nu_G, \tilde{\nu}_G) \le D_T(\mathbf{P}_G, \tilde{\mathbf{P}}_G)$. Substituting this into 51 yields,

$$D_T(\Psi(\mathbf{P}_G), \Psi(\tilde{\mathbf{P}}_G)) \le C_\phi C_{\mathbf{V}} D_T(\mathbf{P}_G, \tilde{\mathbf{P}}_G). \tag{52}$$

Under assumption (H8), $C_\phi C_{\mathbf{V}} < 1$. Therefore, Ψ is a strict contraction on the complete metric space $\mathbf{V}^G_T$. By the Banach Fixed-Point theorem, there exists a unique fixed point $\mathbf{P}^*_G = \Psi(\mathbf{P}^*_G)$. This fixed point guarantees consistency between assumed environment and the realized state-control distribution, thereby establishing the Nash equilibrium. ■

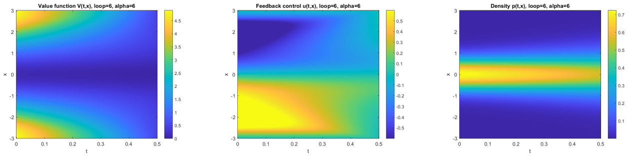
V. A NUMERICAL EXAMPLE

To illustrate the theoretical results, we consider a 1D state and control environment ($n = 1$) that satisfies assumptions (H1)-(H7). For a fixed joint measure $\nu_G(t)$ at time t, denote the induced state network mean field and control network mean field at α by $\mu_\alpha(\nu_G(t)), \theta_\alpha(\nu_G(t))$ respectively. The dynamics and running cost for a generic agent are,

$$f_\alpha(t, x_\alpha, u_\alpha, \nu_G) := (-a \tanh^2(x_\alpha) + bu_\alpha + c\theta_\alpha(\nu_G(t)))$$
$$L_\alpha(t, x_\alpha, u_\alpha, \nu_G) := r_1 u_\alpha^2 + r_2(u_\alpha - \theta_\alpha(\nu_G(t)))^2.$$

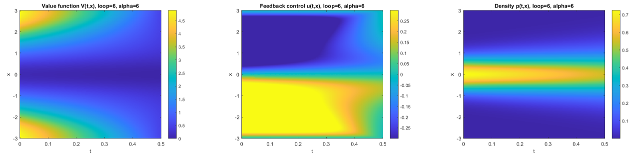
We conduct simulations on two different kinds of constraints. In the first case, we have a universal constraint,

$$U_{\alpha,1}(t, \overline{\lambda}_\alpha) := \Big\{ u \in \mathbb{R} \;\Big|\; u \in [u_{min}, u_{max}] \Big\}.$$



(a) Value function (b) Feedback control (c) State Density

Fig. 1: (a)-(c): Simulations results for a GMFG with time varying control constraints that depends on $\overline{\lambda}_\alpha$



(a) Value function (b) Feedback control (c) State Density

Fig. 2: (a)-(c): Simulations results for a GMFG with an universal control constraints

In the second case, we have a time varying constraint,

$$U_{\alpha,2}(t,\overline{\lambda}_\alpha) \coloneqq \left\{ u \in \mathbb{R} \;\middle|\; u \in [\overline{\lambda}_\alpha - \Delta, \overline{\lambda}_\alpha + \Delta] \right\}.$$

We choose the following parameters,

$$a = 0.8,\ b = 1.0,\ c = 0.6,\ \sigma = 0.45,\ \Delta = 0.3,$$
$$q_1 = 1.0,\ r_1 = 0.6,\ r_2 = 0.8,\ \epsilon = 0.08,\ T = 0.5.$$

We use the following low rank graphon function,

$$\mathcal{W}(\alpha,\beta) = 0.5 + 0.4\cos(2\pi(\alpha - \beta)),\ \alpha,\beta \in [0,1].$$

We propose a numerical scheme and show the implementing result in MATLAB. Select a bounded interval that the state variable concentrated on, and we denote it as $\mathcal{I}_x \coloneqq [x_{min}, x_{max}]$. We discretize that into N_x sample point.

To initiate the iteration, we first find an ensemble of network control mean field $\{\theta_\alpha(\cdot)\}_{\alpha\in[0,1]}$ and solve the HJB equation in that environment. To do that, first solve the point wise optimization problem for the Hamiltonian and integrate backward based on the information on the terminal time or on the next time instant. The terminal cost is set as $g(x) = (1/2)\tanh^2(x)$. The state interval and the time horizon is discretized into N_x and N_t parts respectively.

After the solving the HJB equation, the optimal control law $u^o(t,x)$ is obtained. Substitute this back into the FPK equation with a prescribed initial Gaussian distribution with the mean $\mu = 0$ and the variance $\sigma = 0.55$, we obtain the density flow of both state and control. Consequently, we have the resulting network control mean field.

Iterate the above process and compare the network control mean field $\{\theta_\alpha\}_{\alpha\in[0,1]}$ at each step. The discrepancy of network control mean fields are shown between each iterations. The prescribed maximum iteration is N_{loop}. The graph is discretized into N_α parts.

We choose the following parameters,

$$N_x = 300,\ N_t = 200,\ x_{min} = -3,\ x_{max} = 3$$
$$u_{min} = -0.6,\ u_{max} = 0.6,\ N_\alpha = 12,\ N_{loop} = 6$$

Based on the above parameters, numerical simulations show that the discrepancy between network control mean fields $\{\theta_\alpha(\cdot)\}_{\alpha\in[0,1]}$ in the last two iterations in the two scenarios have infinite norms of 3.83×10^{-3} and 3.83×10^{-5} respectively. In both cases, for some time instant t and some state variable x the optimal control is saturated on the boundaries as it is shown in Fig. 1 and Fig. 2.

VI. CONCLUSIONS AND FUTURE WORKS

This work rigorously establishes the existence of a consistent joint probability measure flow in an Extended Graphon Mean Field Game (EGMFG) subject to control constraints that depend dynamically on the network control mean field. Future research should investigate scenarios where Hamiltonian regularity conditions fail, necessitating the study of viscosity or weak solutions within the EGMFG framework.

Furthermore, exploring restrictive formulations like direct distributional constraints would allow for modeling density limits and congestion penalties. Additionally, transitioning from dense graphons to sparse graph limits via graphexon frameworks [11], [12] will provide a more robust representation of real-world networks with heterogeneous sparsity. Finally, establishing the ϵ-Nash equilibrium property is crucial to rigorously validate these infinite-population strategies when applied to finite-agent systems of size N.

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